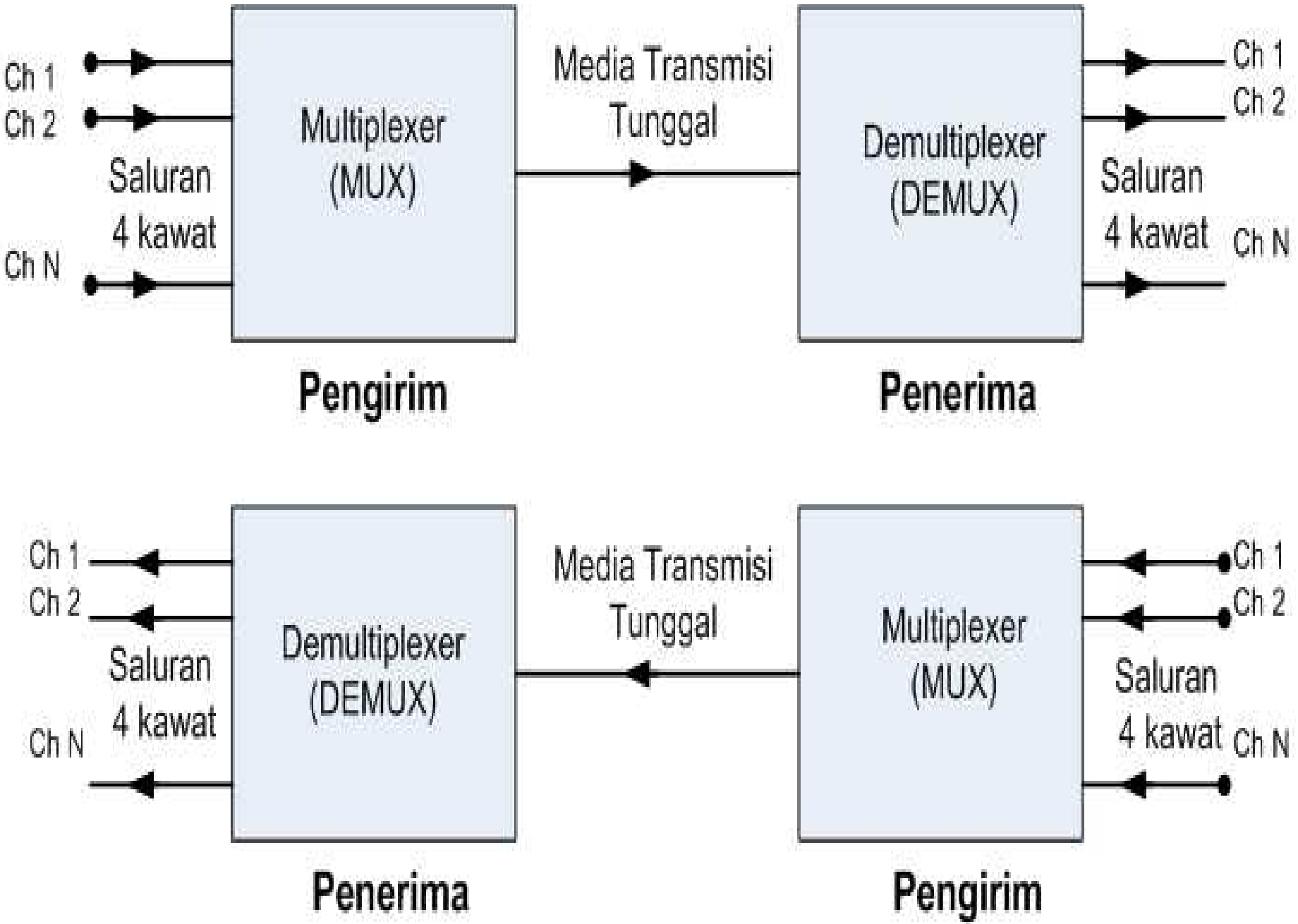
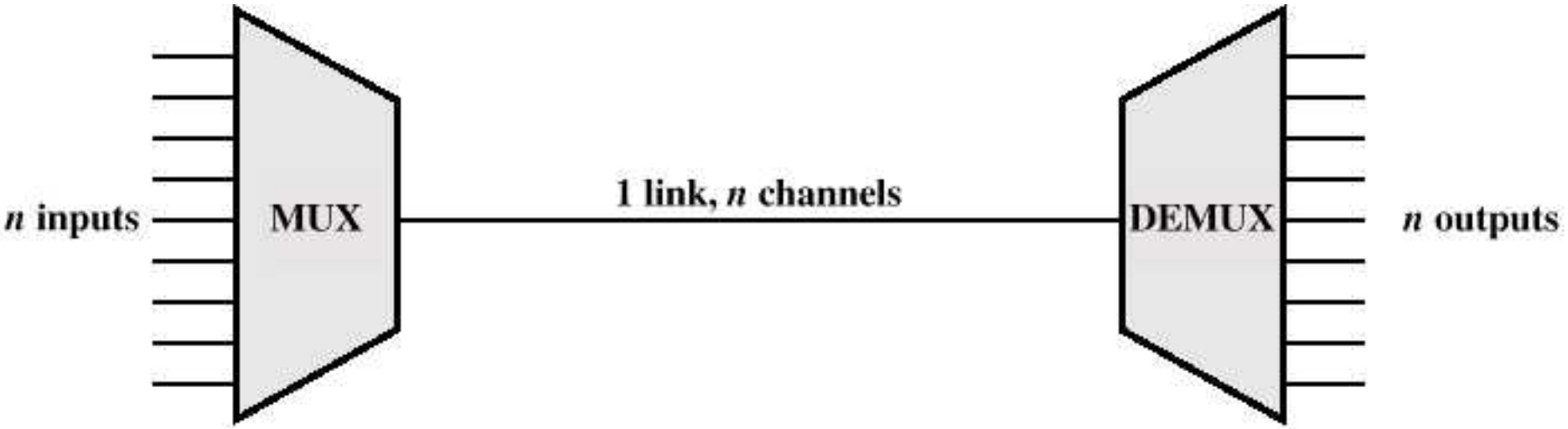
Multiplexing and Demultiplexing (Channelization)

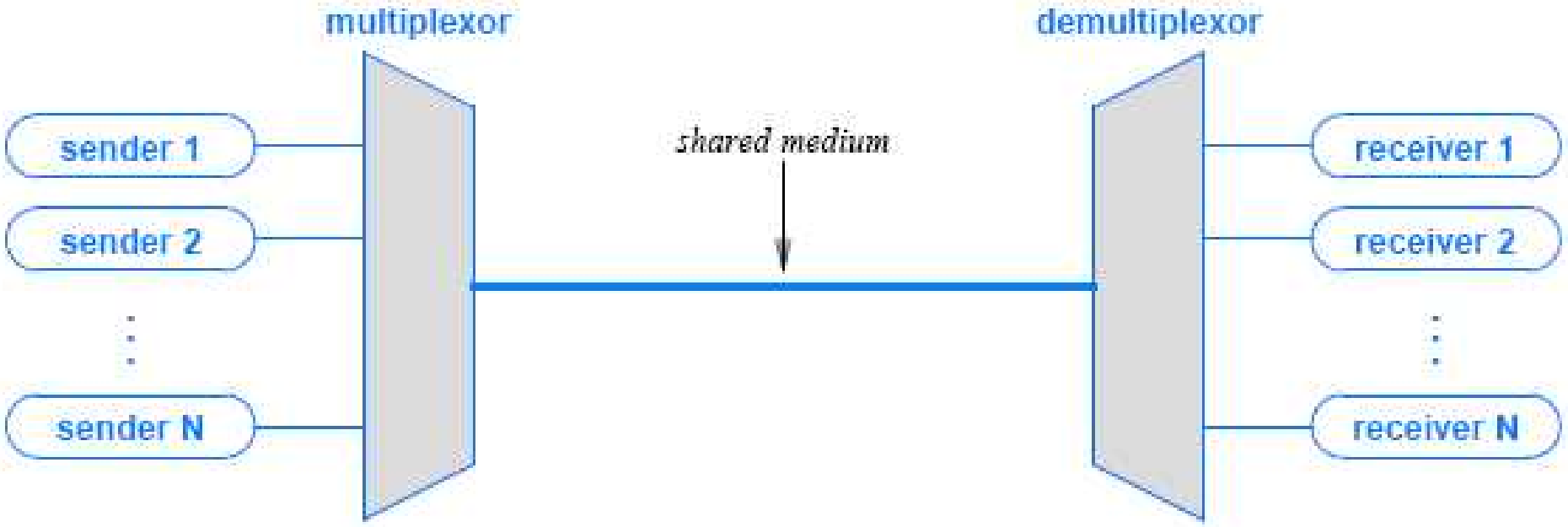
➢ Multiplexing:

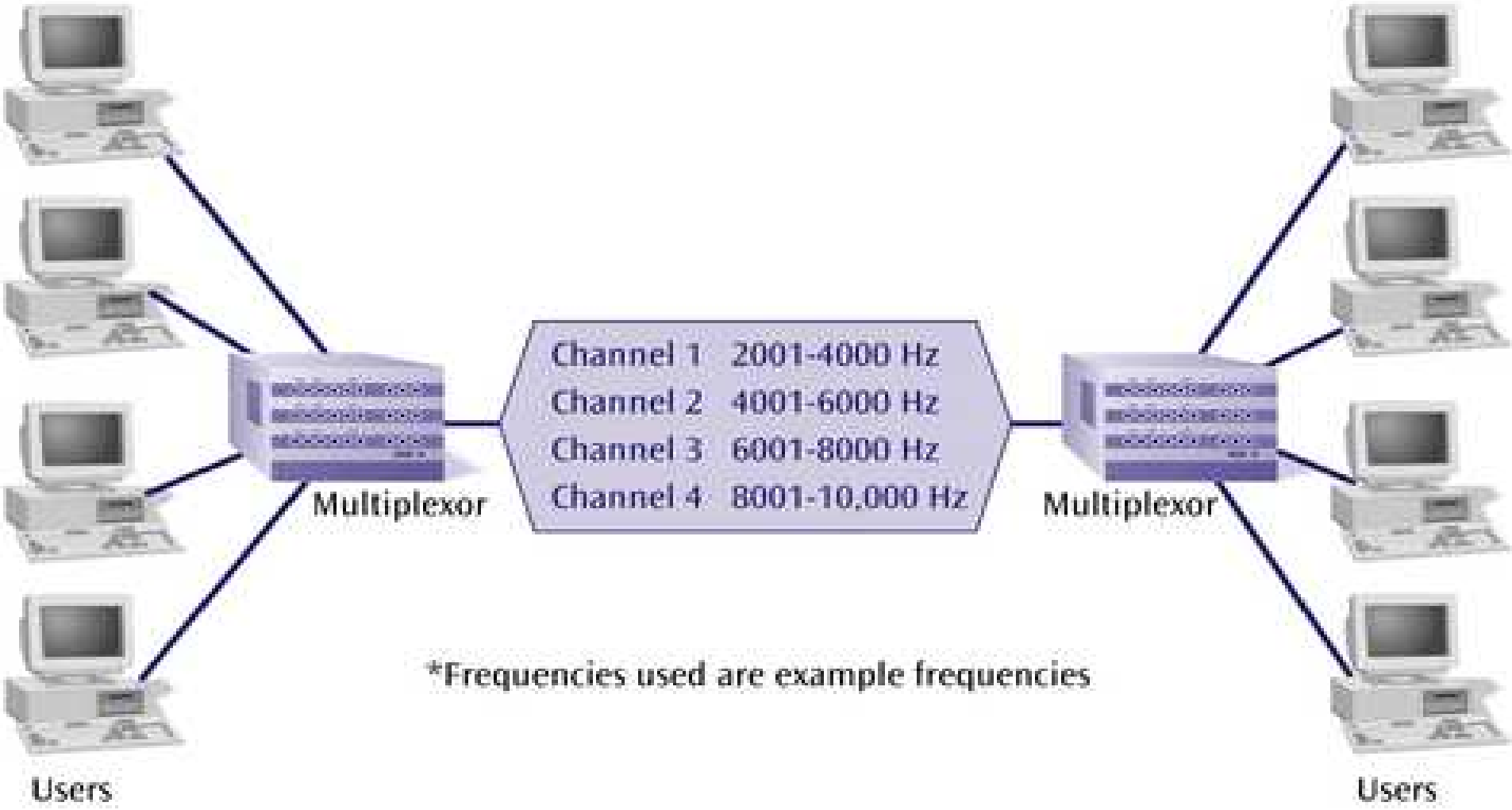
- The **combining of two or more information channels** onto a **common transmission medium.**
- **Two or more** simultaneous **transmissions** on a **single circuit**.
- **Multiplexing:** name given to **techniques** which allow **more than one message** to be transferred via the **same communi-cation channel.**
- **Multiplexing** are **techniques** which allow **k users** to **occupy a channel** for the duration in time, that the channel is available.

- **Multiplexing** to refer to the **combination** of information streams from multiple sources for transmission over a ***shared medium***
- Multiplexing adalah transmisi dua atau lebih kanal (Tele)komunikasi secara serentak melalui satu media transmisi.
- **Multiplexor (MUX)** is a mechanism that implements the concept.
- **Demultiplexing** to refer to the **separation** of a combination back into separate information streams
- **Demultiplexor (DEMUX)** to refer to a mechanism that implements the concept


Ch 1
Ch 2
Saluran
4 kawat
Ch N
Multiplexer
(MUX)
Media Transmisi
Tunggal
Demultiplexer
(DEMUX)
Ch 1
Ch 2
Saluran
4 kawat
Ch N
Pengirim
Penerima
Ch 1
Ch 2
Saluran
4 kawat
Ch N
Demultiplexer
(DEMUX)
Media Transmisi
Tunggal
Multiplexer
(MUX)
Ch 1
Ch 2
Saluran
4 kawat
Ch N
Penerima
Pengirim

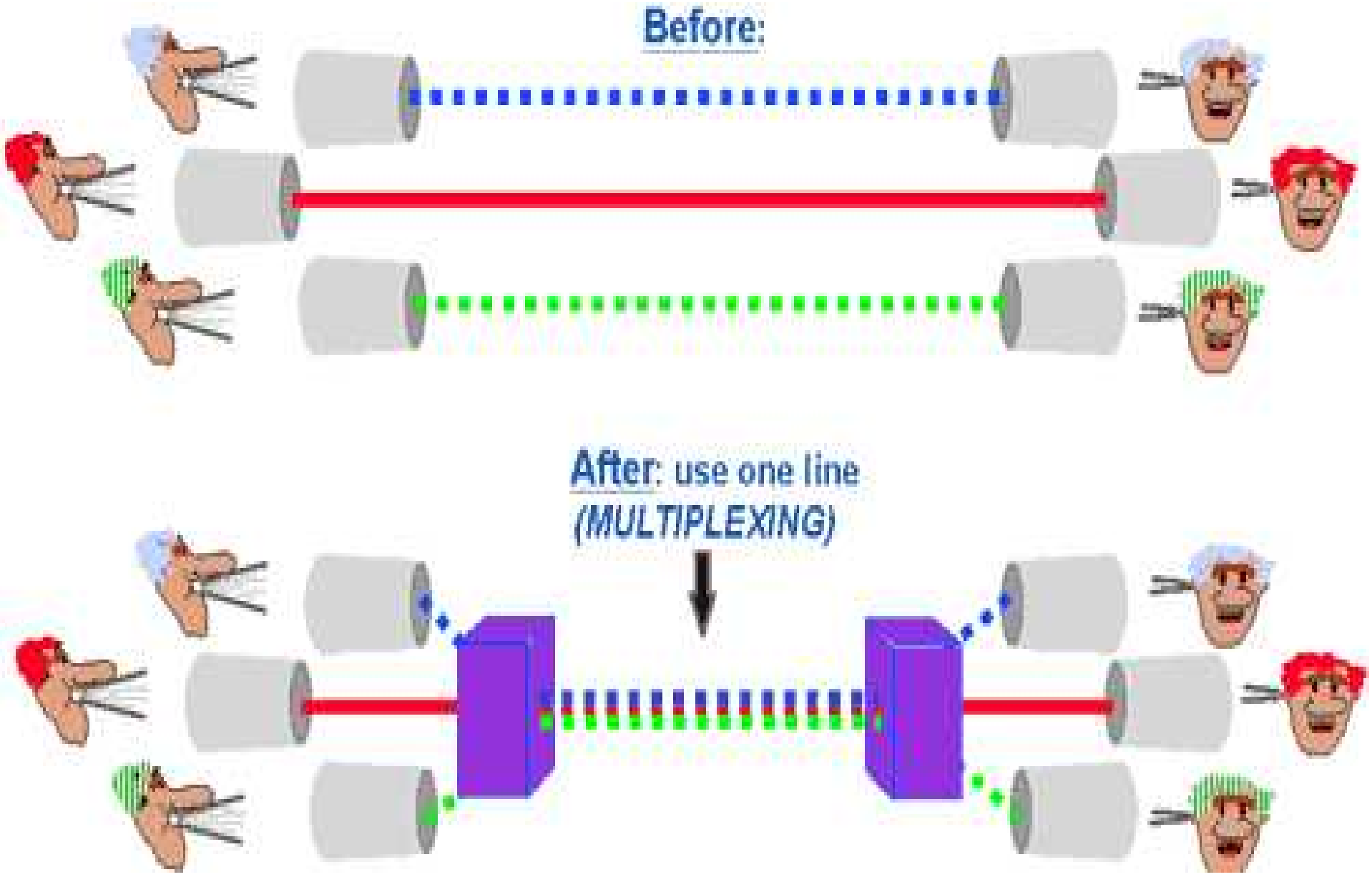

n inputs
MUX
1 link, n channels
DEMUX
n outputs

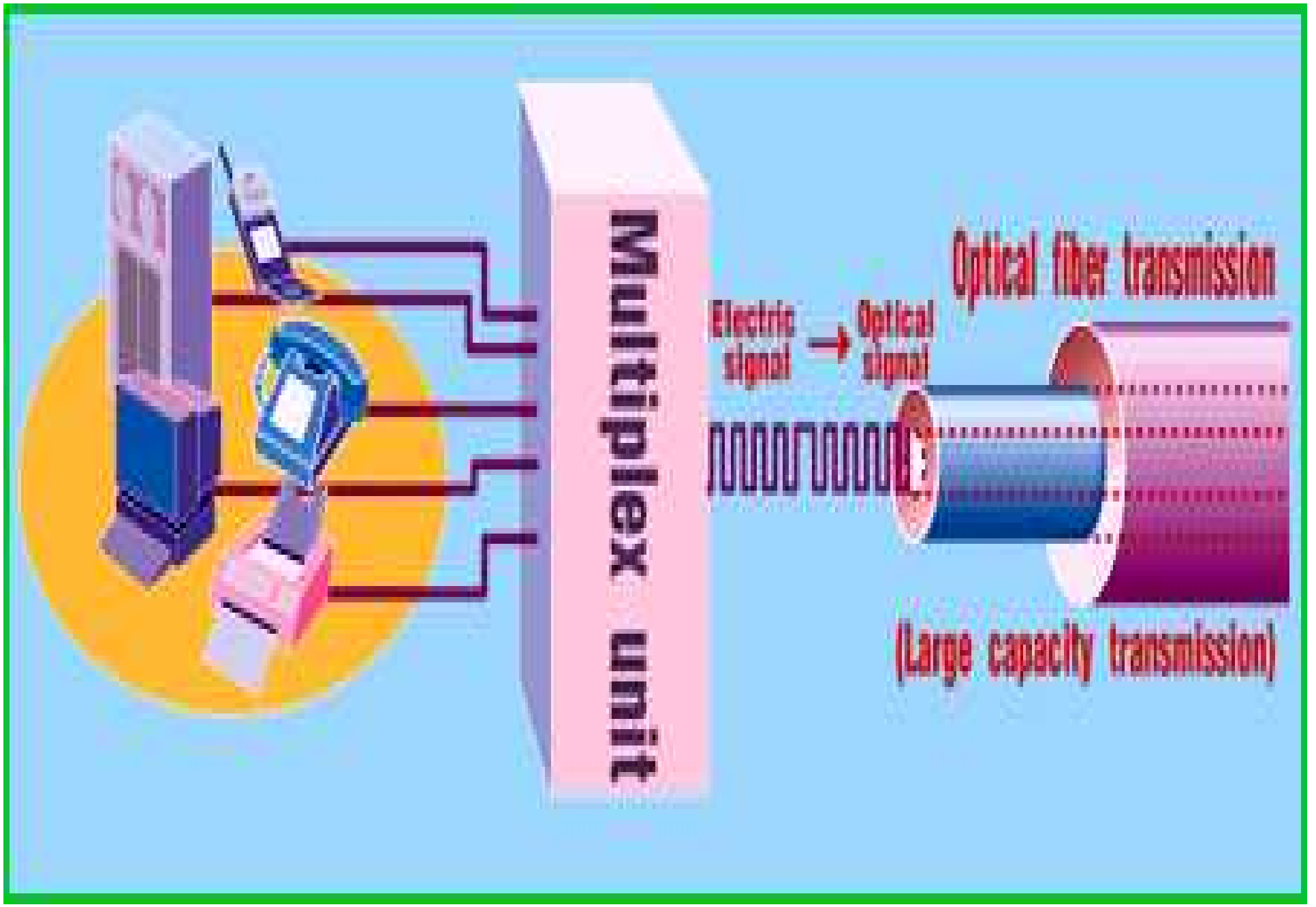

multiplexor
demultiplexor
sender 1
sender 2
sender N
shared medium
receiver 1
receiver 2
receiver N

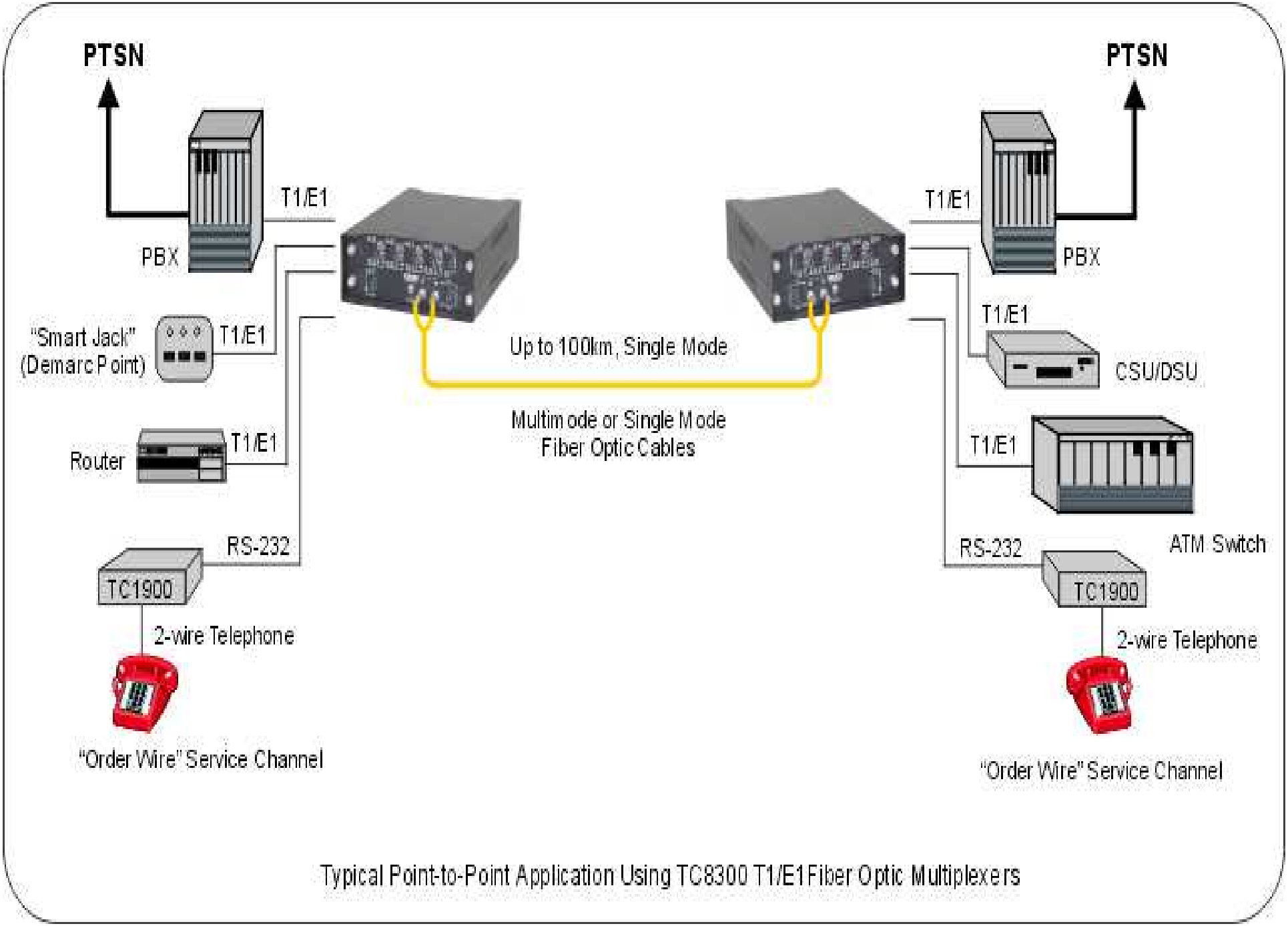

Channel 1 2001-4000 Hz
Channel 2 4001-6000 Hz
Channel 3 6001-8000 Hz
Channel 4 8001-10,000 Hz
Multiplexor
Multiplexor
Users
Users
*Frequencies used are example frequencies

- **Media Transmisi Tunggal (Shared Medium):** could be a transmission line;

- Twisted pair
- Co-axial cable
- A radio system
- Fibre optic cable
- Etc.


Before:
After: use one line
(MULTIPLEXING)


Multiplex unit
Electric signal
Optical signal
Optical fiber transmission
(Large capacity transmission)



Typical Point-to-Point Application Using TC8300 T1/E1 Fiber Optic Multiplexers

- The current techniques that can accomplish this include :
  - ❖ Frequency Division Multiplexing (FDM)
  - ❖ Time Division Multiplexing (TDM)
    - Synchronous Time Division Multiplexing (STDM)
    - Statistical Time Division Multiplexing (STATTDM)
  - ❖ Wavelength Division Multiplexing (WDM)
  - ❖ Code Division Multiplexing (CDM)

- ❑ TDM and FDM are widely used.
- ❑ WDM is a form of FDM used for optical fiber.
- ❑ CDM is a mathematical approach used in cell phone mechanisms.

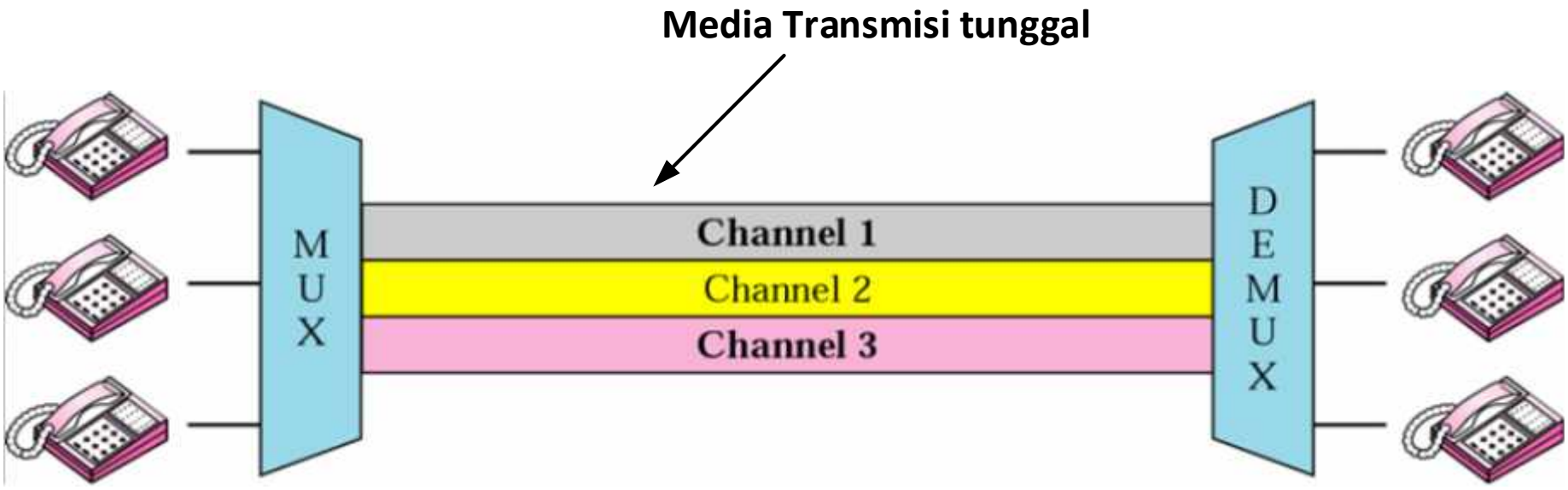
➢Frequency Division Multiplexing (FDM)

- Used in analogue systems
- Analog signaling is used to transmits the Analog signals.
- Since it involves analog signaling, it is more susceptible to noise.
- This technique is the oldest multiplexing technique.
- FDM mechanism will separate the frequency from others without otherwise modifying the signal.
- The deriving of two or more simultaneous, continuous channels from a transmission medium by assigning a separate portion of the available frequency spectrum to each of the individual channels.

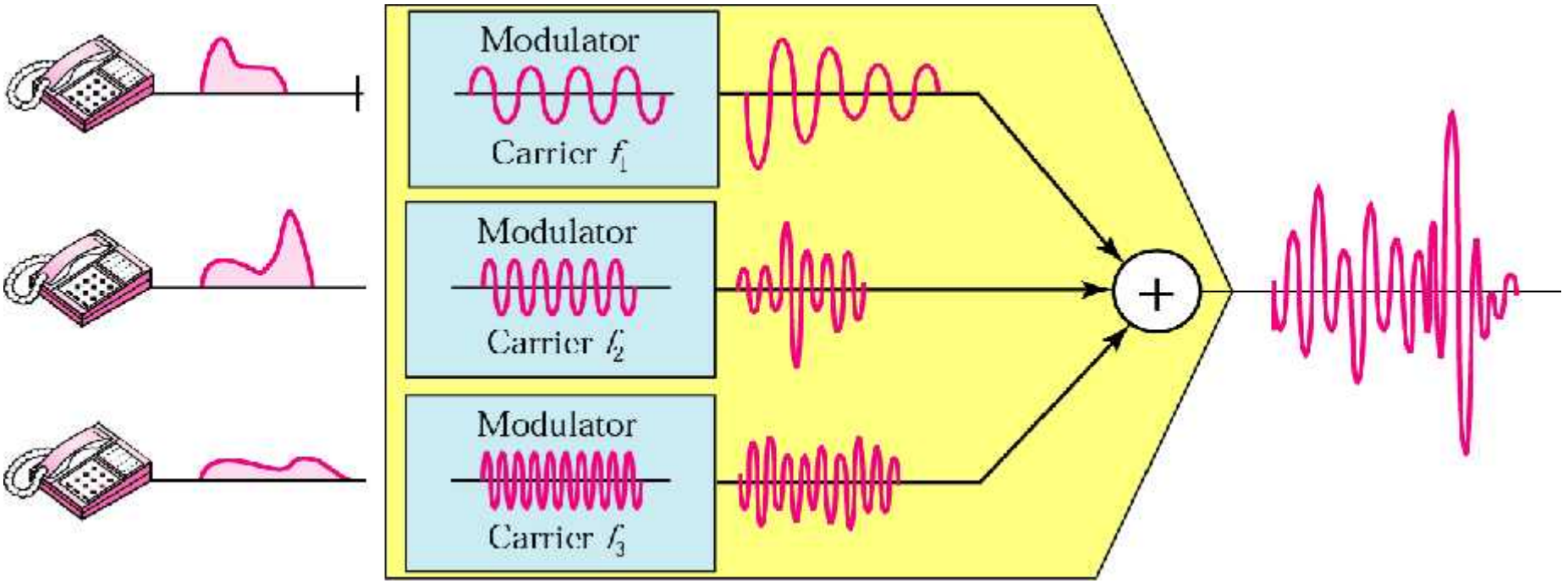
- Transmission is organized in frequency channels, assigned for an exclusive use by a single user at a time.
- If the channel is not in use, it remains idle and cannot be used by others.
- There are channeling frequency plans elaborated to avoid mutual co-channel and adjacent-channel interference among neighboring stations.
- FDMA (frequency-division multiple access): The use of frequency division to provide multiple and simultaneous transmissions.
- Broadcast radio and television, cable television, and the AMPS cellular phone systems use frequency division multiplexing.

FDM

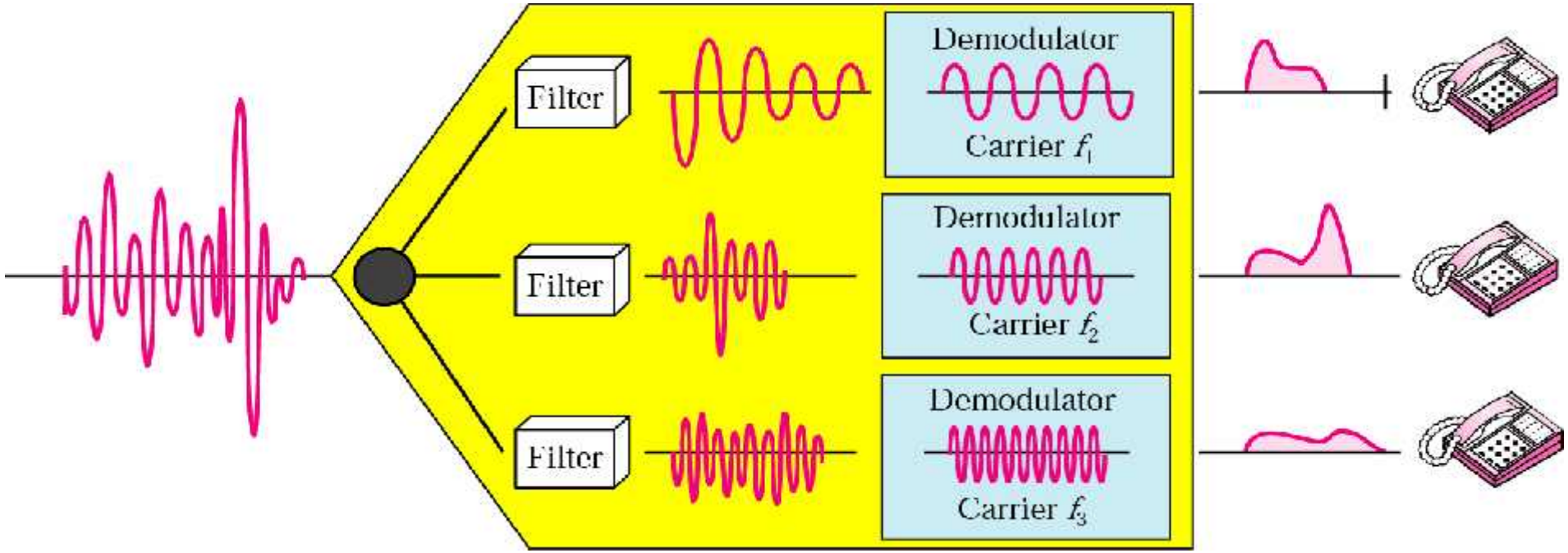
Is the process of translating individual speech circuits (300-3400Hz) into pre assigned slots within the bandwith of transmission medium. and, the preassigned slots are always available to each user



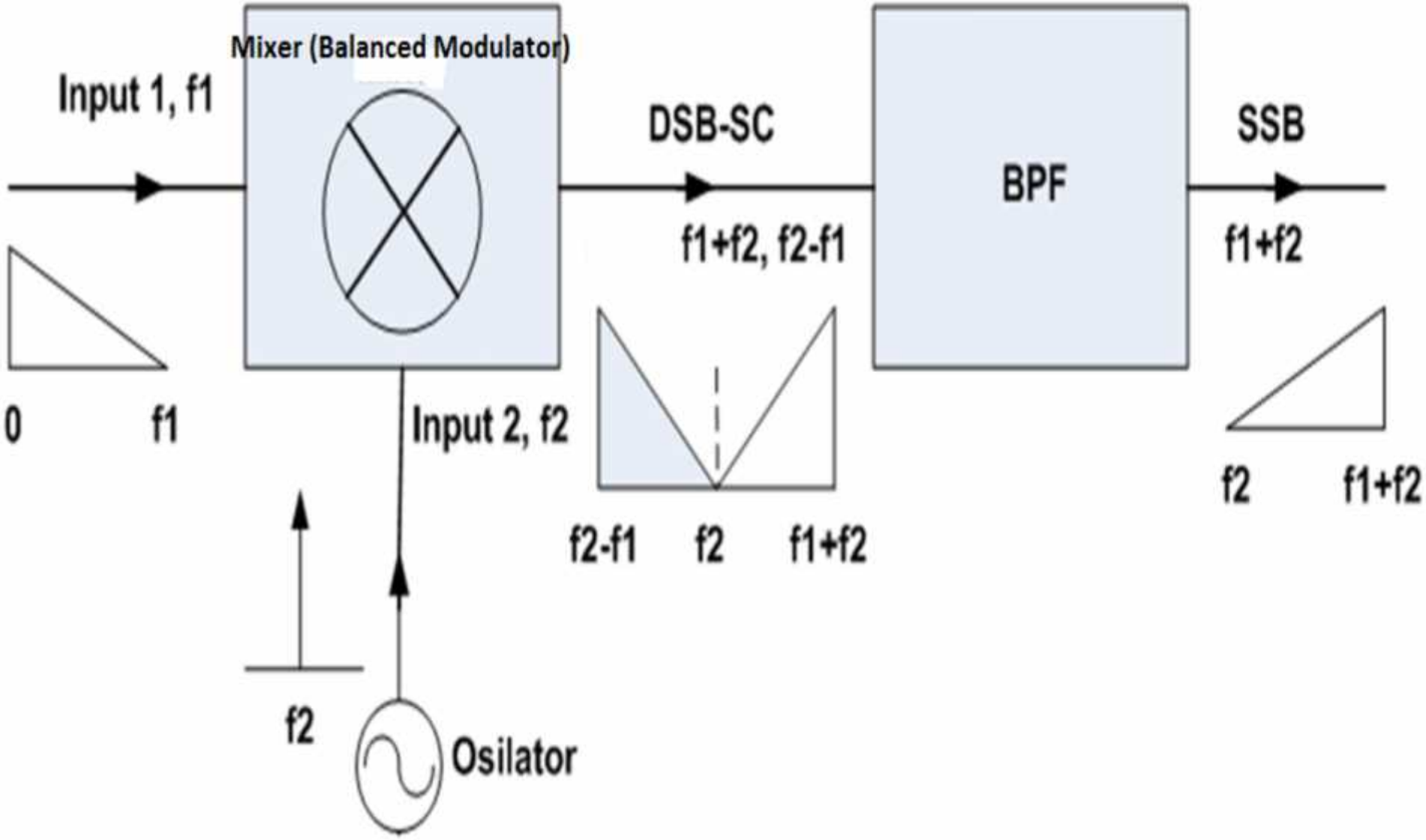
FDM process Multiplexing



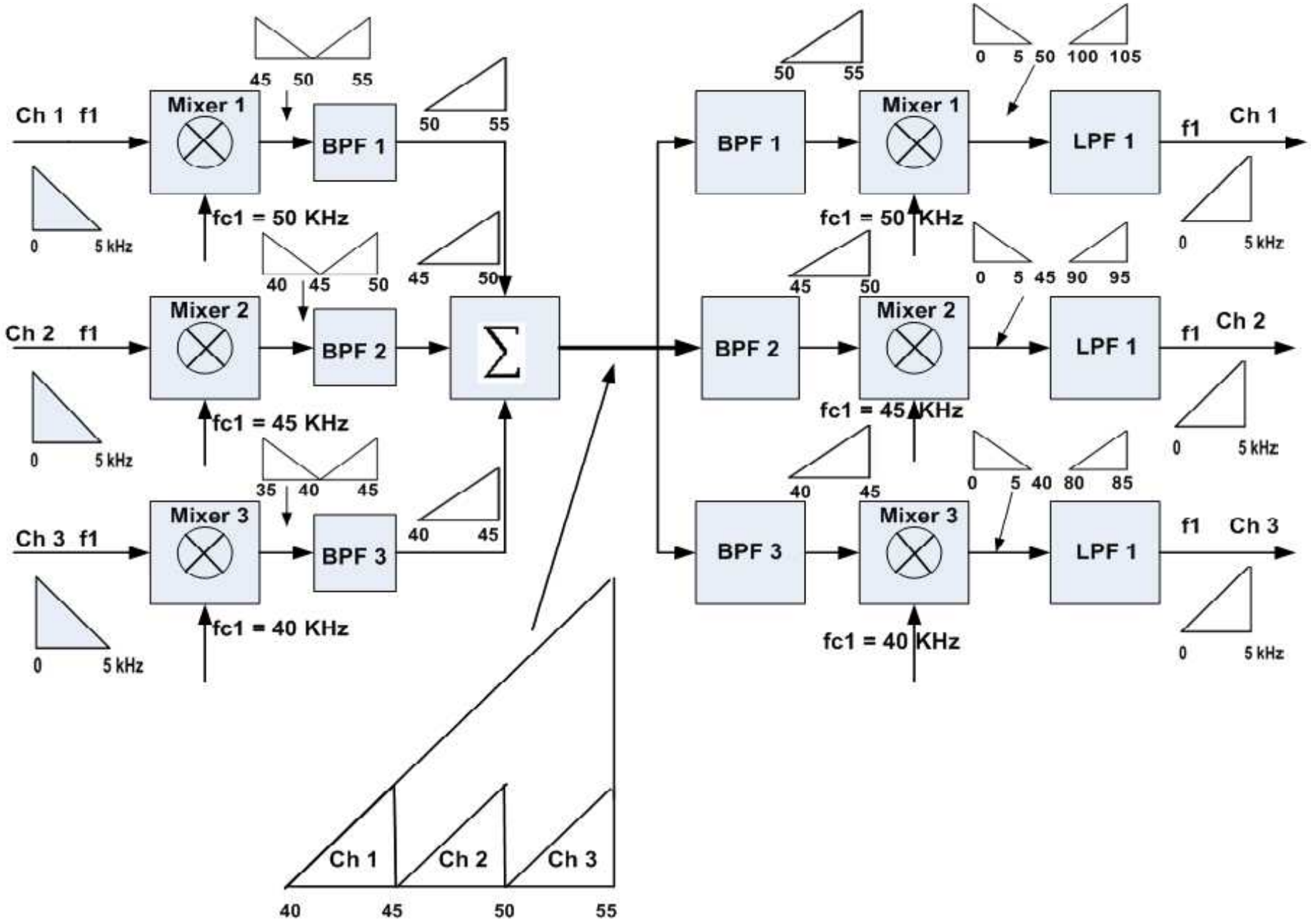
FDM demultiplexing example



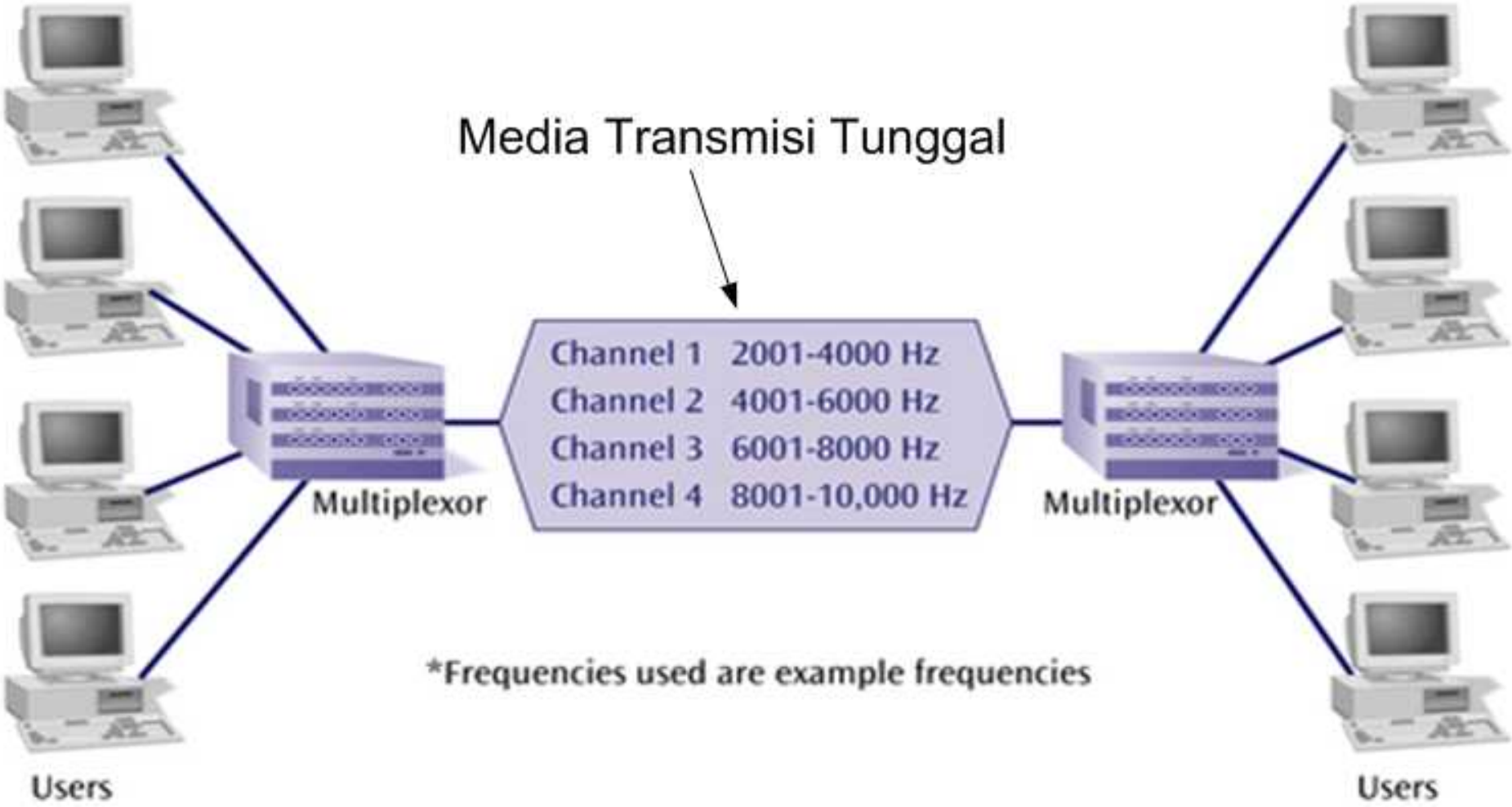
- Prinsip teknik FDM terletak pada teknik pencampuran (mixing) pada suatu mixer (pencampur).



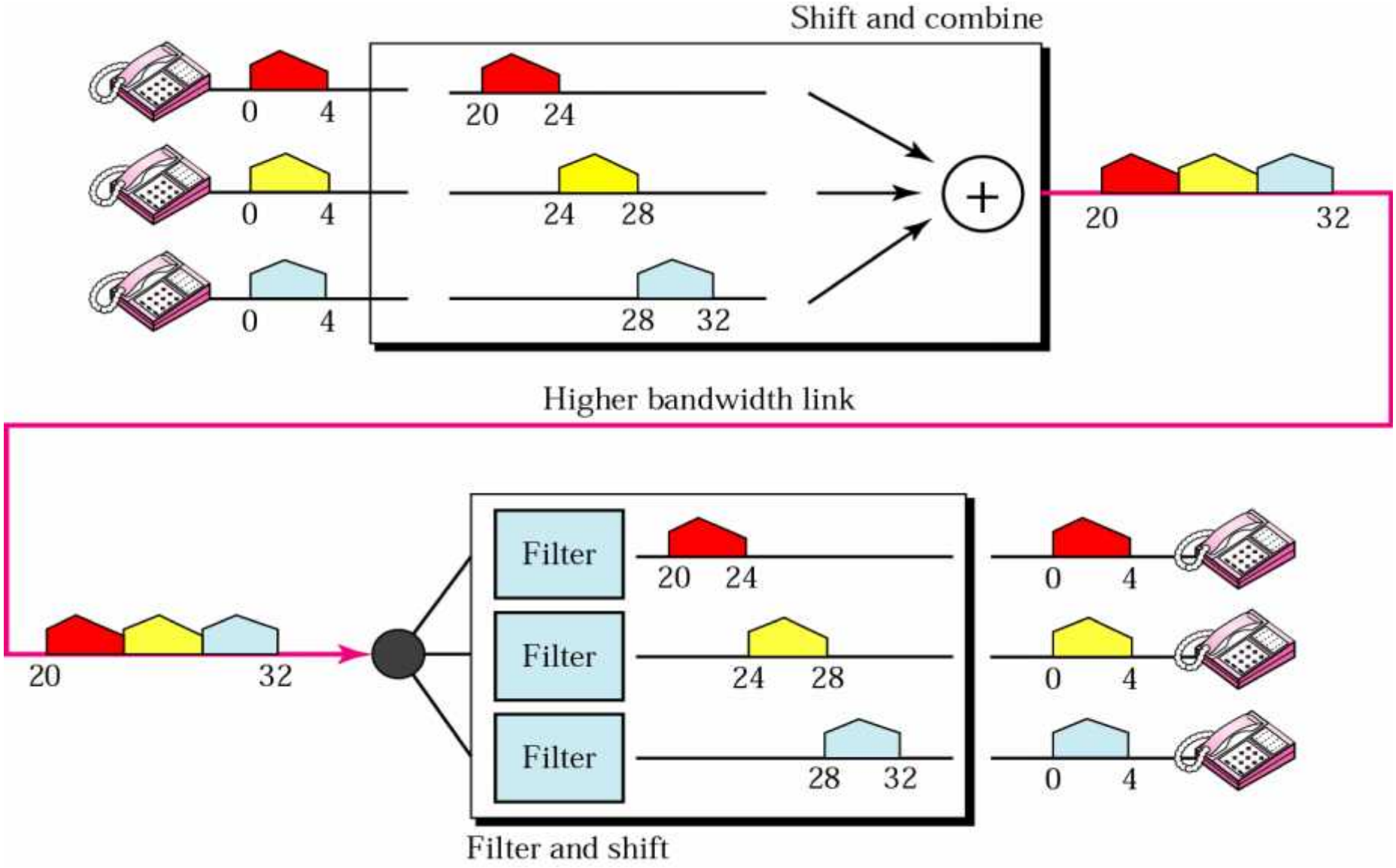
❑ Multiplexing dan Demultiplexing (FDM) Tiga Kanal



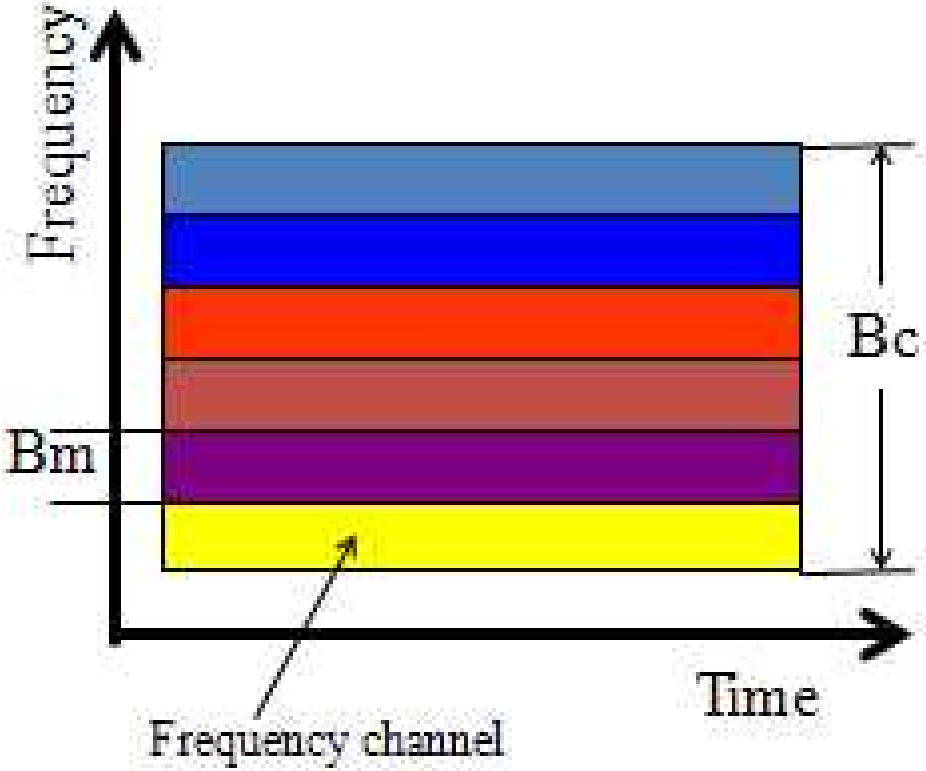
❑ FDM (Frequency Division Multiplexing)

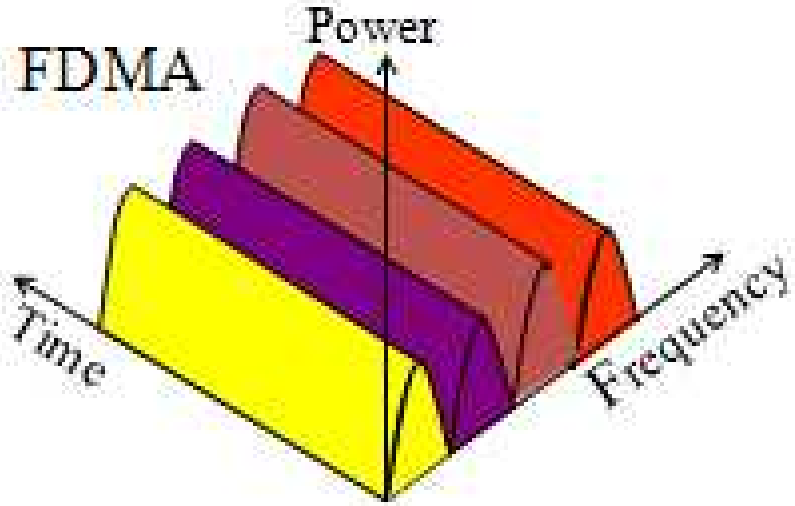


❑ FDM (Frequency Division Multiplexing)



FDM (Frequency Division Multiplexing)



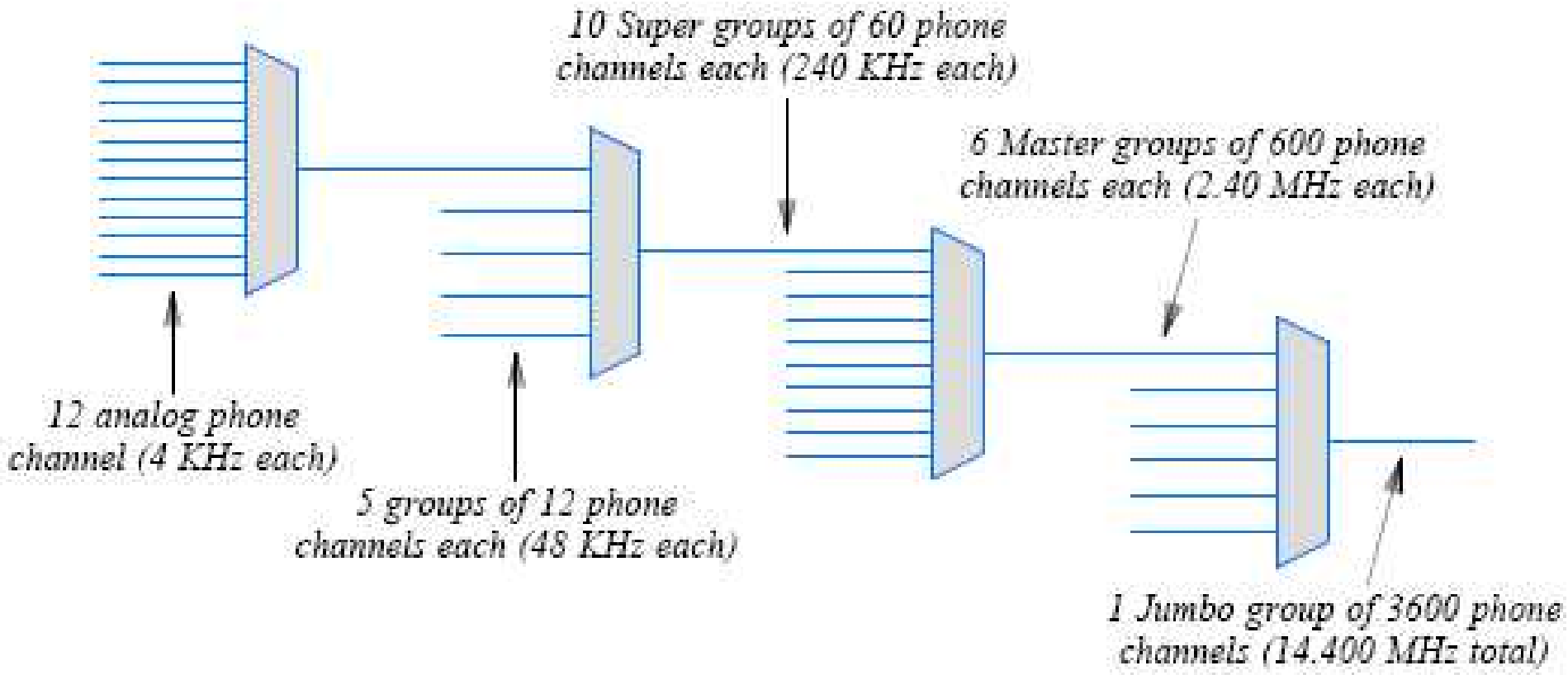


Example: Telephony Bm = 3-9 kHz

❑ Standar CCITT untuk FDM

- ❖ Tingkat satu : Group Standard, memultiplex 12 kanal suara, band frekuensi (60 s/d 108) Khz, bandwidth per kanal sebesar 4 Khz.
- ❖ Tingkat dua : Super Group, menggabungkan 5 Group Standar yang ekivalen dengan memultiplex 60 kanal suara, band frekuensi (312 s/d 552) Khz, bandwidth per kanal 4 Khz.
- ❖ Tingkat tiga : Master Group, menggabungkan 5 Super Group yang ekivalen dengan memultiplex 300 kanal suara band frekuensi (812 s/d 2044) Khz, bandwidth per kanal sebesar 4 Khz.
- ❖ Tingkat empat : Super Master Group, menggabungkan 3 Master group yang memiliki band frekuensi (8516 s/d 12.388) Khz.

➢Hierarchical FDM



Assignment of frequencies for cable television channels

	Channel	Frequency in MHz
Low-Band VHF and Cable	2	54–60
	3	60–66
	4	66–72
	5	76–82
	6	82–88
Mid-Band Cable	95	90–96
	96	96–102
	97	102–108
	98	108–114
	99	114–120
	14	120–126
	15	126–132
	16	132–138
	17	138–144
	18	144–150
	19	150–156
	20	156–162
	21	162–168
	22	168–174
High-Band VHF and Cable	7	174–180
	8	180–186
	9	186–192
	10	192–198
	11	198–204
	12	204–210
	13	210–216

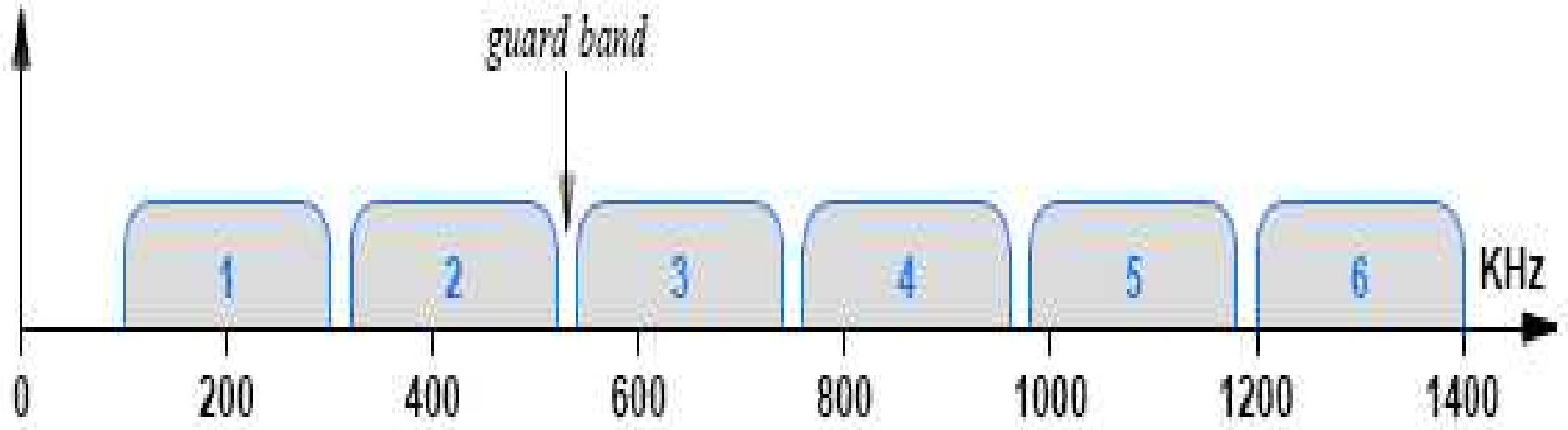
❑ Frequency Division Multiplexing

Channel	Frequencies Used
1	100 KHz - 300 KHz
2	320 KHz - 520 KHz
3	540 KHz - 740 KHz
4	760 KHz - 960 KHz
5	980 KHz - 1180 KHz
6	1200 KHz - 1400 KHz

An example assignment of frequencies to channels with a guard band between adjacent channels.

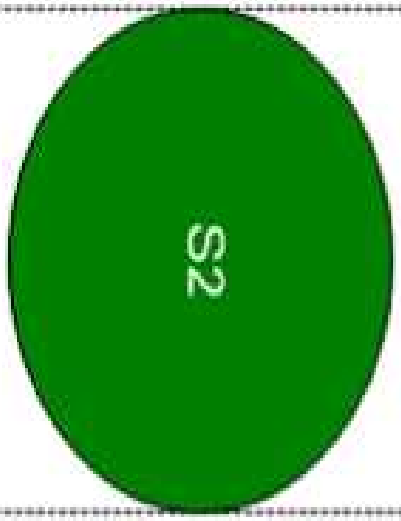
❑ FDM (Frequency Division Multiplexing)

- ❖ To reduce further, the risk of interference place guard space between the frequency spaces
- ❖ Divides the available frequency into non-overlapping bands with guard spaces between to avoid overlapping (adjacent channel interference)

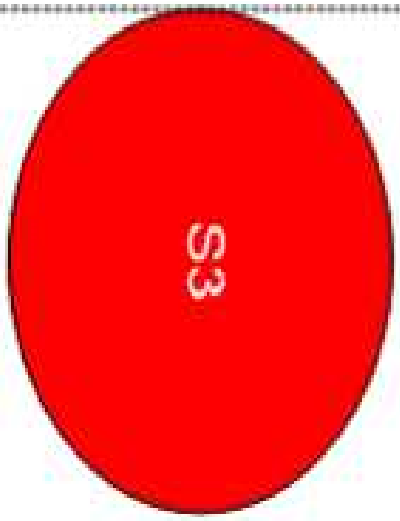


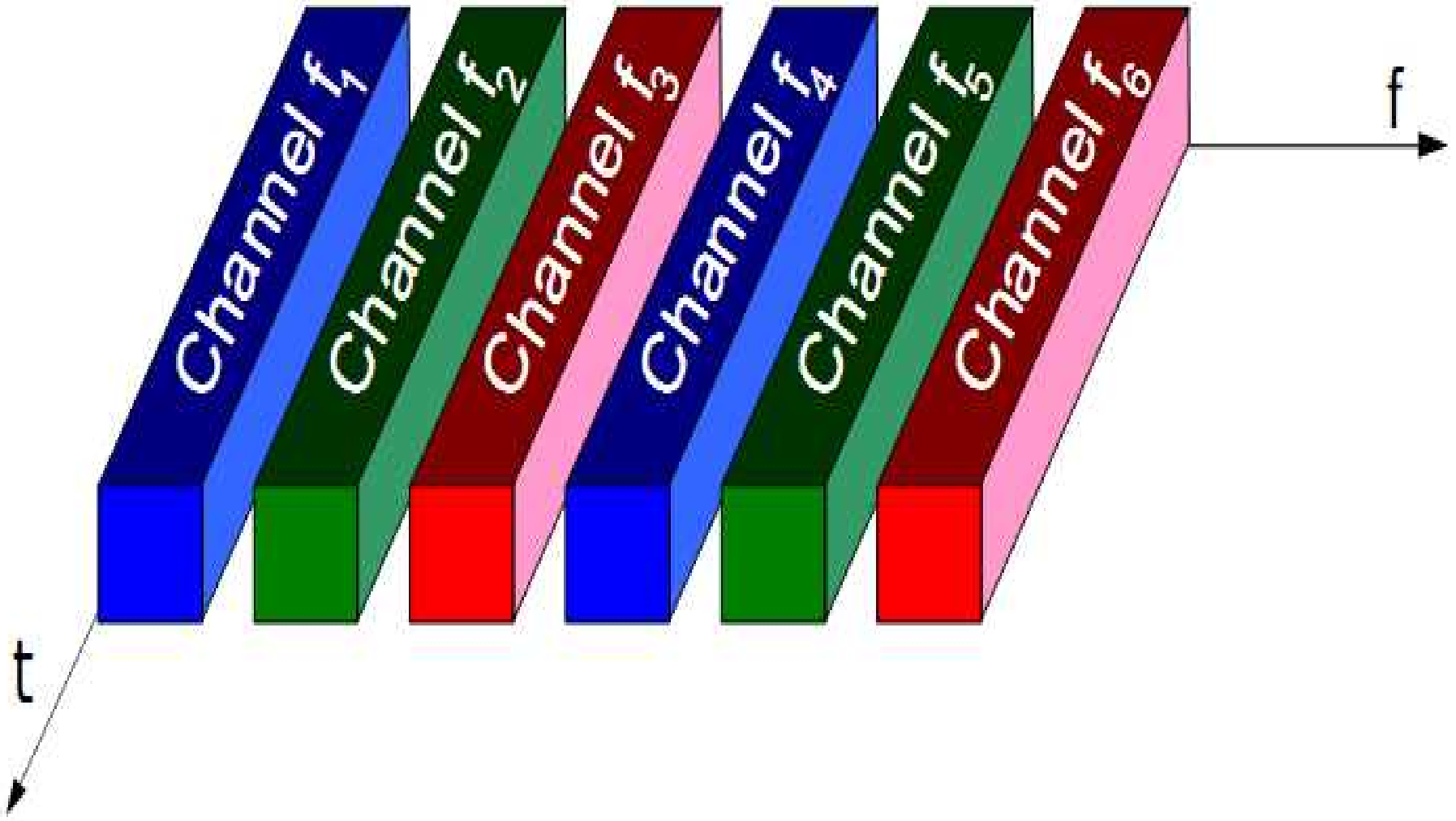


guard space /Guard Band



guard space /Guard Band




Channel f1
Channel f2
Channel f3
Channel f4
Channel f5
Channel f6
f
t